

Advanced (Habanero)

- Q1.** What is the primary purpose of a box-and-whisker plot?
- (A) To display the mean of a dataset
 - (B) To compare the dispersion of two datasets
 - (C) To identify trends in a dataset
 - (D) To display the median and quartiles of a dataset
 - (E) To calculate the standard deviation of a dataset
- Q2.** Which part of the box-and-whisker plot represents the Q_3
- (A) The lower edge of the box
 - (B) The upper edge of the box
 - (C) The median line inside the box
 - (D) The lower whisker
 - (E) The upper whisker
- Q3.** What does the interquartile range (IQR) represent?
- (A) The range of the dataset
 - (B) The difference between the mean and median
 - (C) The spread of the middle 50
 - (C) The standard deviation of the dataset
 - (D) The variance of the dataset
- Q4.** A dataset has $Q_1 = 5$ and $Q_3 = 15$. What is the IQR?
- (A) 5
 - (B) 10
 - (C) 15
 - (D) 20
 - (E) 25
- Q5.** Which of the following is a method to detect outliers in a box-and-whisker plot?
- (A) Checking if data points are within 1 standard deviation of the mean
 - (B) Looking for data points beyond $1.5 \cdot IQR$ away from Q_1 or Q_3
 - (C) Identifying data points that are exactly on the median
 - (D) Determining if data points are within the interquartile range
 - (E) Finding data points with a z -score greater than 2
- Q6.** What is the meaning of the lower whisker in a box-and-whisker plot?
- (A) The minimum value in the dataset
 - (B) The Q_1 value
 - (C) The range of the dataset
 - (D) The value $1.5 \cdot IQR$ below Q_1
 - (E) The median of the dataset
- Q7.** Which of the following statements about box-and-whisker plots is false?
- (A) They can be used to compare the distribution of two datasets
 - (B) They display the mean, median, and mode of a dataset
 - (C) They are useful for identifying outliers
 - (D) They show the interquartile range of a dataset
 - (E) They can be used to display categorical data
- Q8.** A box-and-whisker plot has an upper whisker extending to 20 and $Q_3 = 15$. What is the length of the upper whisker?
- (A) 5
 - (B) 10
 - (C) 15
 - (D) 20
 - (E) 25

Open Ended Questions (FRQ)

- Q9.** Construct a box-and-whisker plot for the dataset: 12, 15, 18, 20, 22, 25, 30, 35, 40, 45, 50. Identify Q_1 , Q_3 , and any outliers.
- Q10.** Compare the distribution of two datasets using box-and-whisker plots. Dataset 1: 10, 12, 15, 18, 20, 22, 25 and Dataset 2: 8, 10, 12, 15, 18, 20, 25, 30. Comment on their central tendency, dispersion, and any outliers.

Chef's Tips

- Tip 1: Ensure students understand the scaling of the box-and-whisker plot.
- Tip 2: Remind students to check for outliers beyond $1.5 \cdot IQR$ from Q_1 and Q_3 .
- Tip 3: Encourage students to compare the distribution of different datasets using box-and-whisker plots.